

JAMES CONLON

📍 Rutherford, NJ ✉ proconlon@proton.me 🌐 github.com/proconlon [linkedin.com/in/proconlon](https://www.linkedin.com/in/proconlon) ☎ 201-600-1119

Education

Boston University

Expected May 2026

B.S. in Computer Engineering

Boston, MA

Minor in Computer Science

Relevant Coursework (GPA: 3.47):

Distributed Systems (Go), Embedded Systems, Operating Systems, Cybersecurity, Client-Server Software

Experience

ei³ : Industrial Internet Intelligence

Jun 2024 – Present

Embedded Software and Security Intern

Pearl River, NY

- Developed a multi-instance OpenVPN solution in Docker (Python, Java, Bash), enabling advanced network routing (fwmark, policy routing, NAT) to secure legacy industrial devices in production.
- Assisted with software rollout on hardware at customer locations by training support staff and documenting the onboarding processes for the product.
- Integrated a containerized remote-access system with certificate management (OpenSSL) and IP allocation databases, ensuring future scalability.
- Applied modern security best practices (Zero Trust, ISO/IEC frameworks) to safeguard older protocols and modernize industrial networks.
- Simulated factory networks (OPC-UA, MODBUS, PLCs), and performed competitive benchmarking against key competitors to evaluate technical capabilities, user experience, and security.

BU Information Services and Technology

Aug 2022 – Dec 2024

IT Support Specialist

Boston, MA

- Diagnosed and resolved a range of hardware and software issues, including Windows/Mac/Linux support, academic software, networking, and identity verification.

Projects

FPGA-Based Keyboard Synthesizer | Verilog, FPGA, Digital Signal Processing, Team Project

Dec 2023

- Developed a digital synthesizer on Artix-7 FPGA, enabling simulation of musical notes and octaves.
- Designed a module for generating PWM signals with variable duty cycles to simulate sound waveforms including square, sine, and triangle waves.
- Integrated PS/2 keyboard input for control, along with a 7-segment display and audio output for feedback.
- Collaborated with team to address challenges such as sound generation and timing issues related to waveform indexing.

Room Occupancy Monitor | Arduino, C++, Onshape, Team Project

Apr 2024

- Collaborated on development of device to monitor the occupancy of a room to prevent overcrowding and create a mechanical barrier to entry.
- Primarily responsible for coding occupancy tracking, integrating various sensors (infrared, ultrasonic) and input/output devices to maximize accuracy and reliability.
- Developed a heuristic algorithm to manage ambiguous sensor data and varying traffic patterns, enhancing the accuracy of room occupancy monitoring.

Android App: Boston Where Are U | Android SDK, Java, Team Project

Dec 2023

- Programmed an Android game for BU students that challenges players to identify local landmarks using a custom built map interface.
- Set deadlines for project development, integrated UI elements and backend logic.
- Implemented key features including custom overlays, TouchImageView map integration, and backend game management with Android navigation components for fragment management.

Technical Skills

Languages: Java, Python, Go, C (systems-level), Bash, Verilog

Networking and Security: Docker, OpenVPN, NAT/VLAN/Routing, PKI/Certificate Management, Wireshark

Tools and Platforms: Linux (Debian), Docker Compose, Android SDK, Kali Linux (for security labs/CTFs), Git